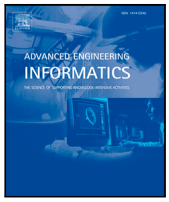
Advanced Engineering Informatics 62 (2024) 102839



Contents lists available at ScienceDirect

Advanced Engineering Informatics

journal homepage: www.elsevier.com/locate/aei



Full length article

GraphSmin: Imbalanced dissolved gas analysis with contrastive dual-channel graph filters

Ke Shu, Huifang Ma *, Jinpeng Yang, Di Zhang

College of Computer Science and Engineering, Northwest Normal University, Lanzhou, Gansu 730070, China

ARTICLE INFO

Keywords:
Dissolved gas analysis
Class imbalanced learning
Graph neural networks
SMOTE
Contrastive learning

ABSTRACT

Dissolved Gas Analysis (DGA) is a widely adopted technique for detecting faults in oil-immersed power transformers. However, in the context of DGA fault diagnosis, the fault class has considerably fewer samples compared to the normal class. Directly constructing fault diagnosis model in such an imbalanced scenario may result in insufficient representation of fault classes, which can ultimately lead to a decrease in diagnostic performance. To tackle this issue, we propose a novel imbalanced DGA model called **GraphSmin**. This approach is equipped with contrastive dual-channel graph filters and deliberately generate minority samples to effectively address the imbalance problem in fault diagnosis. Specifically, similar *K*NN graph (S-*K*NN) and dissimilar *K*NN graph (DS-*K*NN) are established to better reveal the complex relationship between samples. Subsequently, the dual-channel graph filters with contrastive learning is presented to obtain high-quality embeddings of DGA samples. In particular, we expand minority class samples in the embedding space to ensure effective learning of minority class features. Besides, an edge predictor is trained to model the relationship information between nodes. Extensive experiments on two datasets demonstrate the outstanding capability and reliability of the proposed method in imbalanced DGA fault diagnosis. Our code is available at https://github.com/SK-1230/GraphSmin.

1. Introduction

Power transformers play a critical role in power transmission and distribution, as they have a direct impact on the safe and stable operation of power systems. Insulation faults in transformers may result in their shutdown, leading to severe consequences such as compromised reliability of power systems and even power outages [1]. In situations where power outages or system failures occur on a larger scale, the economic losses can be substantial, posing a significant threat to public safety and social stability [2]. Hence, it is crucial to monitor the condition of transformers effectively, implement sound fault diagnosis strategies, and promptly detect and resolve potential faults. These measures are of utmost importance to ensure the reliable and uninterrupted functioning of power systems.

Currently, Dissolved Gas Analysis (DGA) in oil has emerged as a viable method for diagnosing faults in power transformers within the power industry [3]. This technique involves periodically collecting oil samples from power transformers for analysis, aiming to identify gases released from the oil that are indicative of faults. By analyzing the concentration of these fault-related gases, the condition of the transformer can be determined [4]. The diagnostic criteria for transformers typically rely on five common gases: hydrogen (H_2), methane (CH_4), ethane (C_2H_6), ethylene (C_2H_4), and acetylene (C_2H_2) [5,6]. In terms of fault types, the 'Interpretation guidelines for dissolved gas analysis of electrical equipment in operation with mineral insulating oils' (IEC 60599:2007) has reduced the original nine typical faults to six: Partial Discharge (PD), Low-energy Discharge (LD), High-energy Discharge (HD), Low-Temperature overheating (LT), Medium-Temperature overheating (MT), and High-Temperature overheating (HT). Traditional diagnostic methods such as the IEC ratio method, modified ratio method, and Rogers ratio method have been widely employed for transformer fault diagnosis [7]. However, these methods have inherent limitations, including limited applicability and incomplete ratio coding [7,8]. In recent years, researchers have started integrating DGA with intelligent diagnostic methods to enhance the accuracy of transformer fault prediction. Common intelligent diagnostic techniques include Bayesian networks [9], *K*-nearest neighbors (*K*NN) [10], Decision trees (DT) [11], and Support Vector Machine (SVM) models [12] (A detailed description of the aforementioned algorithm can be found in Section 5.1.2). More recently, with the rapid advancements in deep learning, numerous researchers have focused on designing various deep learning-based models for transformer fault diagnosis, yielding notable achievements.

* Corresponding author.
E-mail addresses: 2023212053@nwnu.edu.cn (K. Shu), mahuifang@yeah.net (H. Ma), 202031603201@nwnu.edu.cn (J. Yang), zhangdi@nwnu.edu.cn (D. Zhang).

https://doi.org/10.1016/j.aei.2024.102839
Received 6 May 2024; Received in revised form 5 August 2024; Accepted 14 September 2024
Available online 24 September 2024
1474-0346/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

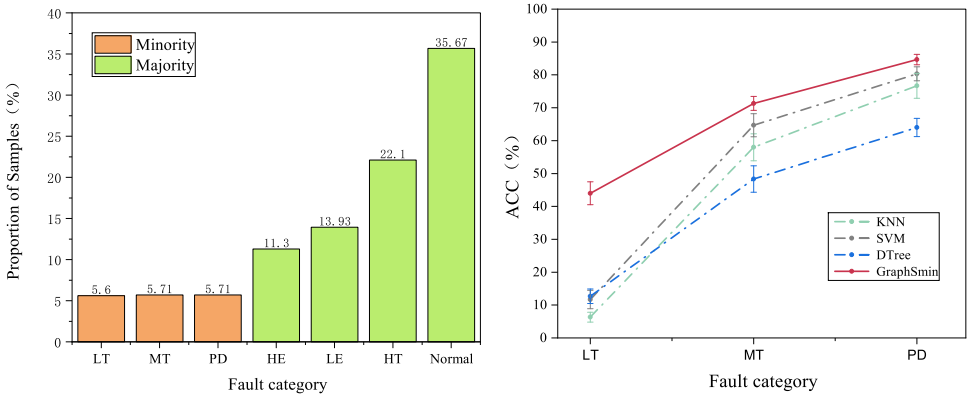


Fig. 1. The distribution of fault categories in the DGA-2000 and the predicted classification accuracy. Fig. 1 (left) illustrate the sample quantities for each fault category in the DGA-2000, which reveals a notable skewed distribution. Particularly, fault categories LT, MT, and PD exhibit significantly fewer samples compared to the others. Learning from a biased sample set can restrict the performance, especially for the minority classes. Fig. 1 (right) presents the predicted classification accuracy of the baseline models and our model on the least three categories in the DGA-2000 under the Long-tailed Sampling setting. Clearly, the proposed model outperforms the baselines significantly in dealing with the identification of imbalanced fault samples.

Table 1
Number of samples for dissolved gas fault data in two types of transformer oils.

Dataset	Fault types							
DGA-200	SD	LT	AD	HT	MT	LMT	PD	*Sum*
	49	19	54	38	9	16	16	201
DGA-2000	PD	LE	HE	LT	MT	HT	Normal	*Sum*
	113	276	224	111	113	438	707	1982

Despite the significant achievements of the aforementioned methods in handling fault diagnosis, we identify two primary limitations that hinder the performance of existing methods. **On the one hand**, existing methods typically treat samples as independent and discrete entities, neglecting potential relationships among samples [13]. In reality, cross-sample learning is crucial for providing effective supervision for rare samples. Thus, exploring the rich and diverse correlations among samples may be beneficial to enhance the accuracy of fault detection. **On the other hand**, once the relationship among samples is established, accurately inferring supervision signals for fault classes presents a new challenge, as redundant information from majority classes can interfere with the representations of fault classes. Therefore, reasonable augmentation of fault class samples is essential for effective fault detection.

To visually depict the data characteristics and the strategies employed to tackle the above limitations in a more intuitive manner, we have conducted a statistical analysis of the relevant dataset and compared the fault detection performance with baseline methods on the three least frequent classes. Fig. 1(a) presents the data distribution of DGA-2000 (with specific statistical information summarized in Table 1 of Section 5), revealing the characteristic of imbalanced distribution of fault data class imbalance—a situation where the number of instances in certain classes significantly exceeds that in others [14]. Indeed, learning from the skewed datasets is of difficult, particularly for the identification and understanding of minority class samples, as majority class samples generally demonstrate higher accuracy [15]. To better understand how the underrepresentation of minority class samples affects model performance in imbalanced settings, we deploy three machine learning frameworks on long-tailed sampling setting (details of the sampling setup can be found in Section 3.1) and analyze the classification accuracy of the three least frequent classes, as shown in Fig. 1(b). Clearly, the limited number of samples fails to provide sufficient representations for effective learning, resulting in suboptimal diagnostic outcomes. In particular, for the least frequent class — Low-Temperature Overheating (LT) — the prediction accuracy of the three baseline methods drops to below 10%.

To this end, in this paper, we propose a GNNs-based data augmentation framework, **GraphSmin**, to tackle the aforementioned challenges by **S**ynthesizing **MIN**ority class samples in the embedding space. Our objective is to develop an efficient graph construction approach that considers sample correlations and seamlessly integrates data augmentation into the graph learning process to derive high-quality node representations for fault detection. Specifically, **to tackle the first challenge**, we utilize both similarity and dissimilarity relationships among features to construct homophilic (similarity) and heterophilic (dissimilarity) graphs of fault samples, capturing diverse correlations within the dataset. **For the second challenge**, we apply the Synthetic Minority Over-sampling Technique (SMOTE) data interpolation algorithm to augment fault samples in the representation space. Besides, we train an edge predictor to model connections between newly generated fault sample nodes and original sample nodes, allowing auxiliary information to propagate along the edges. Moreover, we enhance the learned embeddings by dual-channel graph filters using a contrastive learning mechanism to improve node representations. Extensive experimental evaluations demonstrate the significant performance of our model in effectively handling imbalanced DGA samples.

To summarize, our main contributions can be organized as follows:

- To the best of our knowledge, this work is the first to apply graph machine learning imbalance algorithm to address transformer fault samples diagnosis.
- We propose a novel contrastive dual-channel graph filtering module, which optimizes the embedding representation of fault samples by providing complementary supervisory signals from both homophily and heterophily perspectives.
- We introduce the SMOTE data augmentation algorithm to generate fault samples in the embedding space through data interpolation, thereby providing the model with more supervisory information about minority class. Furthermore, we employ an edge predictor to model the relationships between synthetic samples and original samples.
- Extensive experiments and in-depth analyses show the significant effectiveness of our method in dealing with the imbalance scenarios in dissolved gas analysis in oil. Specifically, compared to baseline models, our approach significantly improves the recognition accuracy of fault samples in imbalanced data.

The rest of this paper is organized as follows: In Section 2, we introduce related work. Section 3 presents the necessary preliminaries and the setup for imbalanced data. Section 4 describes the proposed model in detail, and in Section 5, we validate the effectiveness of our framework with extensive experiments. Finally, in Section 6, we conclude and outline future work.

2. Related work

2.1. Power transformer fault diagnosis

In the power system, the importance of power transformers is self-evident, as their normal operation is closely related to the safe and reliable power supply of the grid. The Dissolved Gas Analysis (DGA) method for analyzing the gas content of insulation oil under different operating conditions provides important basis for evaluating the operation status of transformers [16].

Traditional DGA-based methods, such as the Doernenburg ratio method, the IEC 60599 ratio method, the Roger ratio method, the Duval triangle method, or the Duval pentagon method, are widely used in practice [17]. Although these methods are simple, easy to understand, easy to implement, and widely used by transformer maintenance professionals, they have certain deficiencies in terms of accuracy and uncertainty. In addition, these methods also have the disadvantages of uncertainty in fault severity assessment or erroneous classification in extreme cases [18]. To overcome these shortcomings, researchers have introduced machine learning algorithms such as Bayesian networks [9], Decision trees (DTree) [11], Support Vector Machines (SVM) [12] and K-nearest neighbors (KNN) [10] to explore the potential relationship between dissolved gas fault samples in oil and their corresponding labels.

These methods, while to some extent addressing the DGA problem, unavoidably possess limitations: they excessively focus on the majority class within the model, resulting in under-representation of the minority class. Yet, enhancing the performance of the minority class is crucial, as scarce samples are often more significant. Our model reassesses the problem based on the data imbalance phenomenon existing in actual fault samples, with the aim of achieving superior performance. Specifically, we first employ a dual-channel contrastive learning approach to acquire well-structured embedding representations. Subsequently, we apply the SMOTE algorithm to address the class imbalance issue by oversampling minority class samples. Furthermore, an edge predictor is used to establish connectivity between newly generated samples and the original samples. Ultimately, we feed the enhanced sample information along with structural cues into the classifier to achieve enhanced classification performance.

2.2. Class imbalance algorithms

In various real-world applications of machine learning tasks, class imbalance is a common issue. Due to the significantly large number of samples in the majority class compared to the minority class in the training set, machine learning models often tend to underestimate the minority class, resulting in poor overall classification performance [19]. Most standard machine learning algorithms perform poorly on such datasets because they tend to favor majority class samples, leading to inadequate prediction accuracy for minority classes.

Real-world classification problems are often imbalanced, such as fault detection, medical diagnosis, traffic flow analysis, stock trading, fraud detection and network security [20]. In these scenarios, the number of samples in each class varies significantly. However, learning from skewed datasets is crucial, particularly in identifying minority classes. In recent years, various methods have been proposed to address the issue of class imbalance, which can generally be categorized into two types: data-level methods and algorithm-level methods [21]. Data-level methods aim to create a more balanced learning environment by modifying the feature space or label space of the training data. Basic data-level methods include random over-sampling, random under-sampling, and SMOTE, among others. Algorithm-level methods, on the other hand, focus on modifying machine learning algorithms to handle imbalanced data. These methods typically involve designing and modifying loss functions based on the characteristics of the data to address the imbalance issue.

Our approach falls under the category of data-level methods, as it encourages the model to bias towards minority classes by synthesizing minority class samples in the embedding space. We contemplate synthesizing minority class samples in a latent space that not only encompasses cross-domain information, thereby allowing initially isolated sample points to complement each other with supervised signals, but also provides supplementary information from additional perspectives to achieve superior representational quality. Furthermore, the expanded samples, endowed with the aforementioned advantages, are refined through a meticulously designed edge prediction module. Overall, we comprehensively consider inter-sample complementary signals, contrastive augmentation, and data-level expansion methods to enhance classification performance.

3. Preliminaries

3.1. Notations and imbalance settings

Notations. We use bold uppercase Roman letters to denote matrices, bold uppercase italic letters to represent sets, and bold lowercase Roman letters for vectors. Let $\mathbf{X} \in \mathbb{R}^{N\times d}$ denote the feature matrix of dissolved gas in oil, where N is the number of samples and d is the dimensionality of each feature. Let $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}(|\mathcal{V}| = N)$ represent the set of samples with gas data in the oil. The similar KNN graph and dissimilar KNN graph are denoted as the similar graph $\mathrm{G}^{(S)} = (\mathcal{V}, \mathbf{X}, \mathbf{A}^{(S)})$ and the dissimilar graph $\mathrm{G}^{(NS)} = (\mathcal{V}, \mathbf{X}, \mathbf{A}^{(NS)})$ respectively. $\mathbf{A}^{(S)} = \{0,1\}^{N\times N}$ denotes the adjacency matrix of the similar graph and $\mathbf{A}^{(NS)} = \{0,1\}^{N\times N}$ denotes the adjacency matrix of the dissimilar graph. Taking as an example, If $\mathbf{A}^{(S)}_{i,j} = 1$, it indicates the existence of an edge between nodes v_i and v_j, if $\mathbf{A}^{(S)}_{i,j} = 0$, it implies the absence of an edge between nodes v_i and v_j. For a node v_i, $\mathcal{N}_i$ represents its set of immediate neighbors. Additionally, $\mathbf{A}_{syn} = \tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}$ represents the symmetric normalized adjacency matrix with self-loops added, where $\tilde{\mathbf{D}}$ is the degree matrix, $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ and $\mathbf{I}$ is the n-dimensional identity matrix. Finally, $\mathbf{Y} \in \mathbb{R}^{N\times C}$ represents a one-hot label matrix, where C represents the number of fault categories. $\mathbf{Y}^L$ denotes labeled training samples and $\mathbf{Y}^T$ denotes unlabeled testing samples.

Imbalance Settings. To evaluate our framework under class imbalance scenario, it is necessary to employ certain methods for data partitioning to create class-imbalanced settings. To this end, we adhere to the commonly used class-imbalance setup [14]. Specifically, the test set is obtained via balanced sampling, ensuring an equal number of test samples for each class. For the training set, two imbalance sampling strategies are considered: (1) **Long-tailed Sampling**, which is characterized by an imbalance ratio parameter ρ, resulting in a long-tailed distribution of node counts within each class. Typically, the number of training samples in each class follows an exponential decay, with the decay factor $\rho^{1/C-1}$ ensuring the characteristic long-tail distribution of the data. (2) **Step Sampling.** We designate half of the classes ($C/2$) as minority classes, while the remainder serves as majority classes. The imbalance ratio, denoted by $\rho = max|C_{major}/C_{minor}|$, is defined accordingly. Fig. 2 illustrates the distribution of the training set under these two imbalance sampling settings.

3.2. Graph neural networks

GNNs extract features and perform predictions by learning relationships among nodes on the global structure of the graph [22]. In GNNs, each node updates its representation by interacting with information from its neighboring nodes. This process can be iterated over multiple layers, facilitating information propagation and feature extraction across the entire graph. Typically, they adhere to the "propagation-transformation" paradigm for iteratively updating features:

$$\mathbf{h}_i^{(l)} \leftarrow \mathrm{Trans}(\mathrm{Prop}(\mathbf{h}_j^{(l-1)}; \mathbf{h}_i^{(l-1)})), \forall h_j \in \mathcal{N}_i, \tag{1}$$

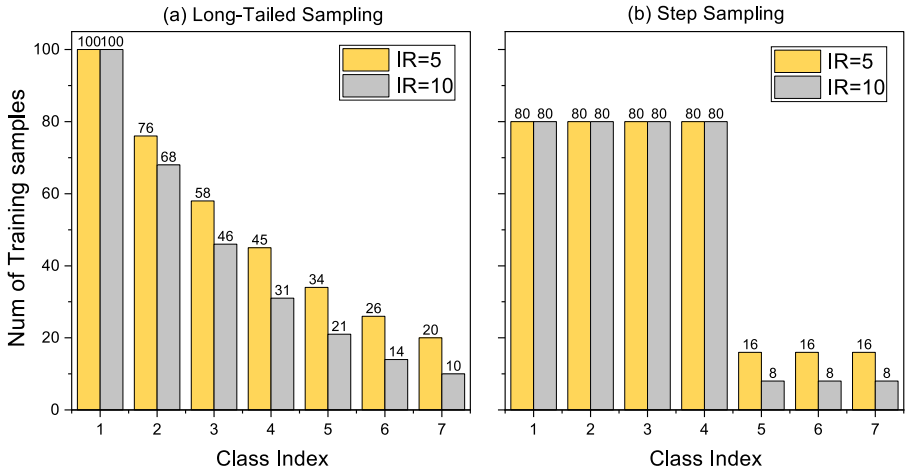


Fig. 2. Illustration of the distribution of each training set in the DGA-2000 dataset under different sampling settings. (a) depicts Long-tailed Sampling, where the majority class is fixed first, followed by progressively removing samples from the minority class to achieve the desired imbalance ratio. (b) employs Step Sampling, where half of the classes are considered minority classes. Similarly, the majority class is fixed initially, followed by pruning the minority class to satisfy the desired imbalance ratio.

where $\mathbf{h}_i^{(l)}$ denotes the node embedding of v_i at the lth layer in graph neural networks, and $\mathcal{N}_i$ represents the neighbors of v_i. For instance, a two-layer GCN [23] can be formalized as:

$$\begin{aligned} &\text{Prop}: \mathbf{z}_i^{(l)} = \mathbf{A}_{syn}\mathbf{h}_i^{(l-1)}, \\ &\text{Trans}: \mathbf{h}_i^{(l)} = \sigma(\mathbf{z}_i^{(l-1)}\mathbf{W}^{(l)}), \end{aligned} \tag{2}$$

where $\mathbf{A}_{syn} = \tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}$ is the symmetric normalized adjacency matrix, $\mathbf{W}^{(l)}$ is the learnable weight matrix at the lth layer, and $\sigma(\cdot)$ denotes the ReLU activation function.

3.3. Problem definition

This work focuses on semi-supervised classification tasks on graph. The objective is to learn a classifier that can effectively identify fault data regarding dissolved gases in oil, particularly the minority class faults. Formally, the DGA fault diagnosis task is defined as follows:

The **input** consists of a feature matrix $\mathbf{X}$, comprising fault data regarding dissolved gases in oil, with partially labeled samples of fault type labels $\mathbf{Y}^L$. The **output** is the predicted values $\mathbf{Y}^T$ of the classifier $\mathcal{F}_{cls}$ for the testing samples.

4. Methodology

In this section, we provide a detailed introduction to the proposed GraphSmin framework, aiming to address the two challenges mentioned earlier. The objective of this framework is to alleviate the adverse effects of under-represented minority class representations by augmenting minority class nodes via data interpolation, while effectively modeling the associations of DGA data. As illustrated in Fig. 3, the framework primarily consists of four components: *(a) Graph Generation; (b) Dual-Channel Representation Learning; (c) SMOTE interpolation; and (d) Augmented Graph Classification*.

Specifically, the similarity matrix of DGA data is computed initially to obtain two views of similarity and dissimilarity among nodes (Fig. 3(a)). In order to provide supervision signals from an additional perspective to enhance the quality of learned node embeddings, an innovative same-different pair-based contrastive learning method is designed (Fig. 3(b) and (e)). After obtaining improved node representations from both views, the representations learned from dual perspectives are fused, and the SMOTE method is utilized for linear interpolation within the minority class to generate new node representations (Fig. 3(c)). Subsequently, we design an edge predictor to establish connectivity between newly generated nodes and original nodes, thereby forming an enhanced graph structure. Finally, the fused node embeddings and the generated graph adjacency matrix are fed together into the classifier to improve the classification performance of the model on minority classes (Fig. 3(d)). Next, we describe each component in detail.

4.1. Constructing fault samples into graph

To capture the potential associations among samples and provide additional supervised signals, we establish a graph from the original data based on the homophily assumption of graph, as adjacent nodes exhibit higher similarity [24]. In practice, when DGA data is converted into a graph representation, node embeddings are affected by both the model updates based on own features of the node and the additional supervisory signals from its neighboring nodes. Leveraging the graph structure enables a more comprehensive integration of relational information between nodes, leading to superior performance in capturing both local and global features of the nodes and, consequently, enhancing the generalization capability of the model.

A common strategy for graph construction is to treat each original data point as a node in the graph, with the feature representation of the original data serving as the feature vector corresponding to the node. Subsequently, the connectivity between nodes is determined by computing the similarity among node pairs. Specifically, for a node v_i, the similarity relationship between v_i and the rest of the nodes is computed to obtain the similarity vector set $s_i = \{s_{i,j} | j = 1, 2, 3, \ldots, N\}$, where $s_{i,j}$ represents the similarity between nodes v_i and v_j. Next, the elements of s_i are sorted to obtain the sorted similarity vector set $\tilde{s}_i$, such that $\tilde{s}_{i,j} > \tilde{s}_{i,k}$ when $j < k$. Following the same approach, the similarity vector for each node is computed to obtain the final sorted similarity matrix $\tilde{\mathbf{S}} \in \mathbb{R}^{N\times N}$, where N is the number of nodes. The sorted similarity matrix $\tilde{\mathbf{S}}$ is used to construct the KNN graph. In our work, we employ the Gaussian Kernel Function for computing the similarity between node pairs. Given nodes v_i and v_j, their similarity can be computed as follows:

$$s_{i,j} = e^{-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{\tau}}, \tag{3}$$

where $s_{i,j}$ denotes the similarity value between sample v_i and sample v_j, $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{x}_j \in \mathbb{R}^d$ represent the original feature of nodes v_i and v_i respectively, d represents the dimension of the original features of the nodes, and τ is the temperature hyperparameter of the Gaussian Kernel Function.

Based on the obtained sorted similarity matrix $\tilde{\mathbf{S}}$, for each node, the top K nodes are selected as the most similar neighbor set $\mathcal{N}_p$ to construct the adjacency matrix of the similar KNN graph, where

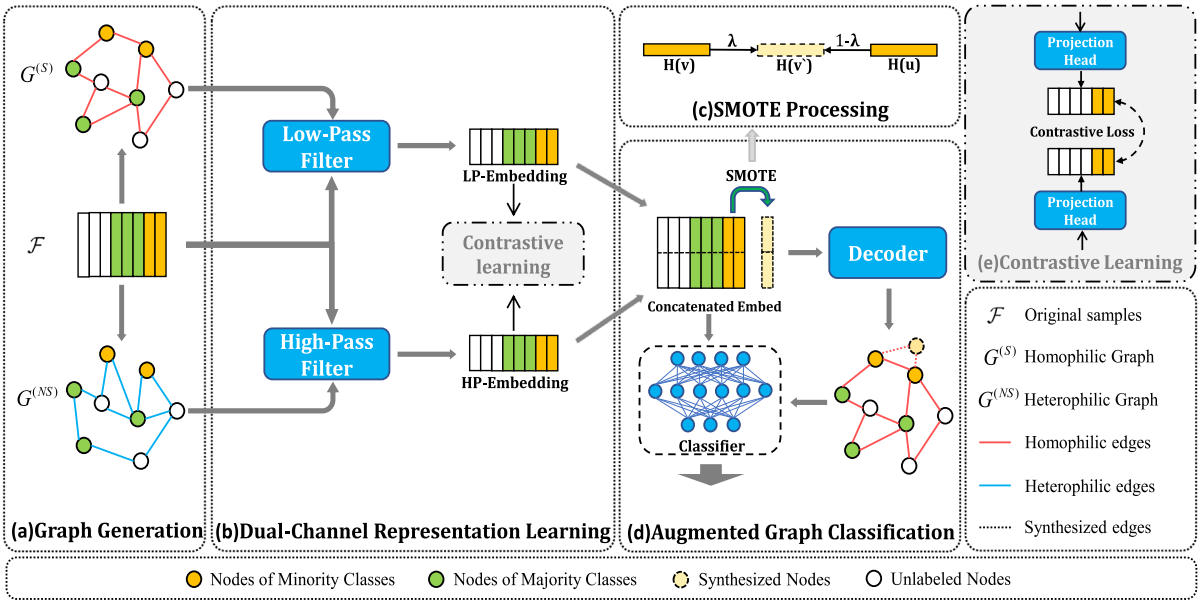


Fig. 3. GraphSmin overview. Orange nodes denotes minority class and green is majority class. (a) depicts the graph generation process, where two views of the graph are generated using the KNN algorithm. After obtaining the two graphs, they, along with the original fault sample features, are fed into two filters (b) to obtain their respective embedding representations. Contrastive learning (e) is then employed to optimize the consistency between the corresponding information from the two views, where the embedding obtained from the high-pass filter is utilized to provide additional information supplementation. Subsequently, the model merges the two embedding representations. (c) represent the synthesize process of fault samples with SMOTE algorithm. Finally, we establish the graph structure relationship between the synthesized samples and the original dataset in (d), and the augmented graph is inputted into the classifier for model prediction.

$\left|\mathcal{N}_p\right| = K$. The adjacency matrix $\mathbf{A}^{(S)}$ corresponding to $\mathbf{G}^{(S)}$ is defined as:

$$\mathrm{A}_{i,j}^{(S)} = \begin{cases} 1, & v_j \in \mathcal{N}_p \\ 0, & \text{others.} \end{cases} \tag{4}$$

Taking node v_i as an example, if the constructed adjacency matrix $\mathrm{A}_{i,j}^{(S)} = 1$, it indicates that node v_j belongs to the set $\mathcal{N}_p$. Thus, the obtained graph structure information will be utilized to capture the interrelationships among nodes. Note that using different hyperparameter values of K to obtain the adjacency matrix will to some extent influence the final results of the model, as the structural information of the graph has a certain impact on the performance of model. In Section 5.4 (hyperparameter analysis), we will discuss the selection of the K.

Although the above method captures the connections between samples to a certain extent, considering only the similar relationships may lead to suboptimal results. The reasons are as follows: (1) It is prone to overfitting. The aggregation update paradigm of GNNs smooths the representations of neighboring nodes, making the features of nodes close in distance tend to be consistent, which will cause the model to excessively learn a large amount of similar information; (2) Insufficient captured information. The KNN graph construction approach considers only local pairwise similarities between nodes, which may not fully capture the complex relationships and dependencies between samples. As a result, the obtained embeddings may not adequately represent the underlying structure and patterns in the data. Therefore, we consider data augmentation methods on the graph, namely graph contrastive learning (GCL) [25], to enhance the quality of node embeddings learned by the model.

Inspired by the work of Liu et al. [26], we construct sample connections not only from a similarity perspective but also from a dissimilarity perspective. We anticipate that through GNN learning, the representations of adjacent nodes in the similar KNN graph will converge, while those in the dissimilar KNN graph will diverge. In other words, we expect that node representations obtained through contrastive learning methods will achieve enhanced intra-class cohesion and inter-class separation.

Specifically, based on the obtained sorted similarity matrix $\tilde{\mathbf{S}}$, for each node, we select the last K nodes as the least similar neighbors set $\mathcal{N}_q$ to construct the dissimilar KNN graph $\mathbf{G}^{(NS)}$, where $\left|\mathcal{N}_q\right| = K$. The final adjacency matrix representation is denoted as $\mathbf{A}^{(NS)}$, It is worth mentioning that while $\mathbf{G}^{(NS)}$ provides additional information as the dissimilar relationship, the primary relationships of the model still rely on the similar KNN graph. We illustrate a simple process example of constructing the KNN graph in Fig. 4. In correspondence with the notion of similarity relations, the dissimilarity adjacency matrix $\mathbf{A}^{(NS)}$ is defined as:

$$\mathrm{A}_{i,j}^{(NS)} = \begin{cases} 1, & v_j \in \mathcal{N}_q \\ 0, & \text{others.} \end{cases} \tag{5}$$

Learning representations of fault samples on a constructed KNN graph offers substantial advantages. **Firstly**, the KNN graph effectively captures local structures within the data by connecting each node to its K-nearest neighbors, which aids the model in learning the importance of nodes within their local context. This is preferable to directly learning from fault samples, as the KNN graph can directly leverage the similarity between nodes. **Secondly**, GNNs, when applied to KNN graphs, integrate both local features and global neighborhood information efficiently. They have exhibited enhanced robustness against complex patterns and noise, and provide additional neighborhood supervision when sample sizes are limited. **Lastly**, by constructing a dual-view approach, when the information learned from one view is insufficient for generalizing to unseen data, the alternate view offers supplementary information to improve model generalization. Therefore, constructing a KNN graph and utilizing GNNs for feature learning from fault samples is highly beneficial.

4.2. Dual-channel contrastive learning

Although we have captured both homophilic and heterophilic relationship between samples, directly merging the information of the same node under homophilic and heterophilic perspective is insufficient to obtain good embedding representations. GCL as an unsupervised learning method on graphs, attempts to maximize the mutual information

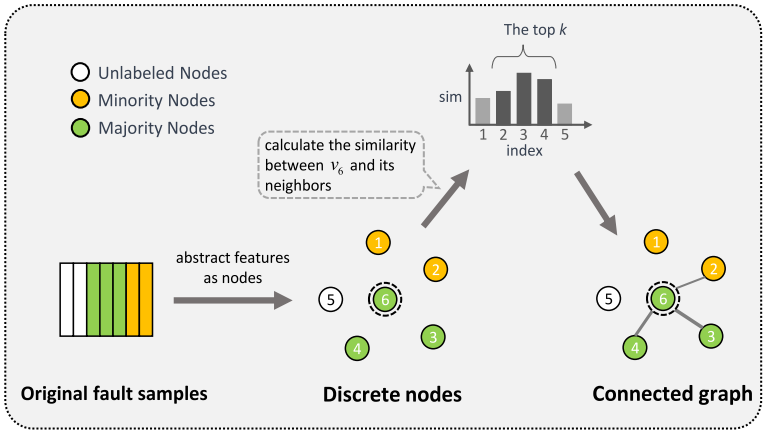


Fig. 4. The illustration of constructing a *K*NN graph using v_6 as an example. We illustrate the process of constructing a similar *K*NN graph using v_6 as an example. First, we calculate the similarity between v_6 and all nodes in the graph. Then, we select the top K nodes with the highest similarity values as neighbors of v_6. Finally, edges are formed between v_6 and the selected neighboring nodes. Similarly, the construction of the dissimilar *K*NN graph involves selecting the top K nodes with the lowest similarity values.

(MI) [27] between inputs and their representations by contrasting positive and negative sampled counterparts, thereby enhancing the quality of nodes. Therefore, we first learn node representations under homophilic and heterophilic perspectives, and then enhance the node representation information under both perspectives through GCL. In the ablation experiments in Section 5.3, we validate the effectiveness of the contrastive learning module.

Dual-Channel Encoder. To learn node embedding representations from the $\mathbf{G}^{(S)}$ and $\mathbf{G}^{(NS)}$, we introduce two different GNNs as encoders to perform low-pass and high-pass filtering, respectively. In this study, we select the representative method GCN [23] and GraphSage [28], as the backbone to construct the low-pass and high-pass filters. We argue that different GNNs may have different strategies for aggregating surrounding information, thereby affecting the final results.

Specifically, for the homophilic perspective, the aggregation of conventional GNNs can be viewed as a low-pass graph filtering process, where the operation of GNNs smooths the representations of neighboring nodes, resulting in closer nodes having more similar embedding representations [29]. For node v_i, its embedding is updated by aggregating information from neighbors as follows:

$$\mathbf{H}_0^{(S)} = \mathbf{X}, \mathbf{H}_l^{(S)} = \sigma(\mathbf{A}_{syn}^{(S)}\mathbf{H}_{l-1}^{(S)}\mathbf{W}_{l-1}^{(S)}), \tag{6}$$

where $\mathbf{X}$ represents the original features of nodes, We consider $\mathbf{X}$ as the input $\mathbf{H}_0^{(S)}$ for the 0th layer of the low-pass filter, $\mathbf{H}_l^{(S)}$ represents the embedded representation after the lth layer convolution, $\sigma(\cdot)$ represents the ReLU activation function, $\mathbf{A}_{syn}^{(S)} = \tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}$ is the symmetric normalized Laplacian matrix, $\mathbf{W}_{l-1}^{(S)}$ represents the weight matrix of the lth layer, $l \in \{1, 2, 3, \ldots, L\}$ denotes the number of convolutional layers, The output $\mathbf{H}_l^{(S)}$ of the low-pass filter is eventually taken as the learned node embedding representation $\mathbf{H}^{(S)}$ under the homophilic perspective.

Unlike in the homophilic view where neighboring nodes tend to have more similar node representations, the edges in the heterophilic graph connect nodes of different types, i.e., neighboring nodes have lower similarity. Directly applying the smoothing operation of the low-pass filter can blur the corresponding node information, leading to reduced discriminability of nodes [30]. To better utilize the heterophilic edge relationships, we employ high-pass graph filtering, the inverse operation of low-pass filtering, to sharpen node features along edges and retain high-frequency graph signals [31]. The process of obtaining the high-pass filter output is as follows:

$$\mathbf{H}_0^{(NS)} = \mathbf{X}, \mathbf{H}_l^{(NS)} = \sigma(\mathbf{R}^{(S)}\mathbf{H}_{l-1}^{(NS)}\mathbf{W}_{l-1}^{(NS)}), \tag{7}$$

where $\mathbf{R}^{(NS)} = \mathbf{I} - \mu\mathbf{A}_{syn}^{(NS)}$ is the symmetric normalized Laplacian matrix, and μ serves as a hyperparameter responsible for controlling the strength of the high-pass filter. Finally, the output $\mathbf{H}_l^{(NS)}$ of the high-pass filter is taken as the learned node embedding representation $\mathbf{H}^{(NS)}$ under the homophilic perspective.

Graph Contrastive Learning. After obtaining the node embedding representations under two perspectives, we enforce consistency constraints using graph contrastive learning. The training process of graph contrastive learning is based on the InfoNCE principle [32], aiming to maximize the mutual information between positive pairs while minimizing the mutual information between negative pairs to obtain effective node representations. We employ a classical graph contrastive learning method [33] to enhance the node representations obtained under both perspectives.

Specifically, for a given node v_i, the embedding $\mathbf{h}_i^{(S)}$ generated under the homophily perspective (*K*NN similarity graph) is considered as the anchor, while the embedding $\mathbf{h}_i^{(NS)}$ generated under the heterophily perspective (*K*NN dissimilarity graph) is taken as the positive sample. The embeddings of other nodes in both perspectives are naturally treated as negative samples [34]. Formally, we define the critic $\theta(\mathbf{h}_i^{(S)}, \mathbf{h}_i^{(NS)}) = \mathrm{Sim}(f(\mathbf{h}_i^{(S)}), f(\mathbf{h}_i^{(NS)}))$, where $\mathrm{Sim}(\cdot,\cdot)$ represents cosine similarity, and f is a nonlinear projection employed to enhance the expressive power of critic. This projection f is realized through a two-layer MLP. For each positive sample pair $(\mathbf{h}_i^{(S)}, \mathbf{h}_i^{(NS)})$, we define the pairwise objective as follows:

$$\mathcal{L}\left(\mathbf{h}_i^{(S)}, \mathbf{h}_i^{(NS)}\right) = -\log \frac{e^{\theta\left(\mathbf{h}_i^{(S)}, \mathbf{h}_i^{(NS)}\right)/\tau}}{\underbrace{e^{\theta\left(\mathbf{h}_i^{(S)}, \mathbf{h}_i^{(NS)}\right)/\tau}}_{\text{the positive pair}} + \underbrace{\sum_{k\neq i} e^{\theta\left(\mathbf{h}_i^{(S)}, \mathbf{h}_k^{(NS)}\right)/\tau}}_{\text{inter-view negative pairs}} + \underbrace{\sum_{k\neq i} e^{\theta\left(\mathbf{h}_i^{(S)}, \mathbf{h}_k^{(S)}\right)/\tau}}_{\text{intra-view negative pairs}}}, \tag{8}$$

where τ is the temperature hyperparameter for contrastive learning. Note that we follow the work of Zhu et al. [33], where the positive samples are the embeddings of the same node under a different view, and the negative samples come from two sources: inter-view nodes and intra-view nodes, corresponding to the second and third terms in the denominator, respectively. Since the two views are symmetric, the loss $\mathcal{L}\left(\mathbf{h}_i^{(NS)}, \mathbf{h}_i^{(S)}\right)$ under the other view is defined similarly. The overall objective function is defined as the average of all positive sample pairs, formally expressed as:

$$\mathcal{L}_{gcl} = \frac{1}{2N}\sum_{i=1}^{N}(\mathcal{L}(\mathbf{h}_i^{(S)}, \mathbf{h}_i^{(NS)}) + \mathcal{L}(\mathbf{h}_i^{(NS)}, \mathbf{h}_i^{(S)})). \tag{9}$$

Furthermore, we illustrate the contrastive learning mechanism in Fig. 5 to provide a clearer description of this process.

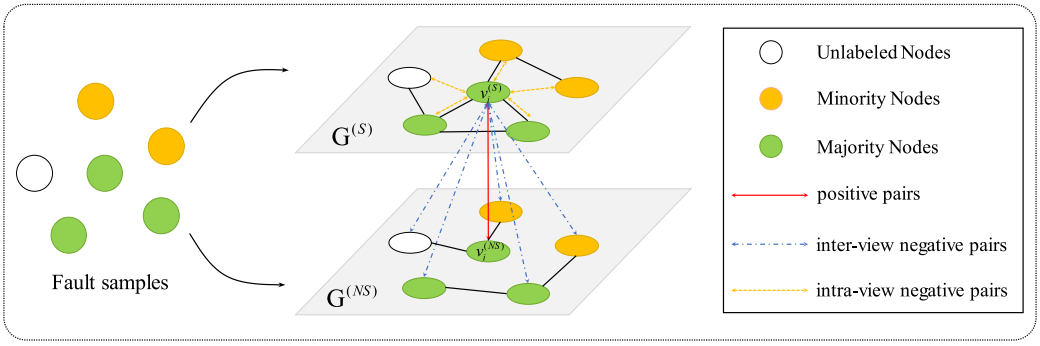


Fig. 5. Illustration of the contrastive learning implementation process using v_i as an example.

Through contrastive learning, we enhance the node representations corresponding to the two perspectives. Subsequently, we integrate the learned representations from both perspectives to construct the final node embedding matrix $\mathbf{H} = \beta\mathbf{H}^{(S)} + (1-\beta)\mathbf{H}^{(NS)}$, where β is used to adjust the weighting of embeddings from different views. The fused embeddings preserve both homophilic and heterophilic relationships among nodes in the graph structure, causing the embedding representations of node pairs within the same class to become more similar, while those of node pairs from different classes gradually move apart.

4.3. SMOTE and synthesizing edges

SMOTE. After obtaining node embeddings in the feature space, we apply augmentation to generate synthetic nodes for minority classes. The Synthetic Minority Over-sampling Technique (SMOTE) is widely acknowledged as the benchmark method for tackling class imbalance in machine learning. Its popularity stems from its straightforward implementation and effectiveness across diverse problem domains [35]. Therefore, we employ the well-established SMOTE algorithm in our approach. The core of the SMOTE technique involves creating new samples by interpolating between existing minority class instances. By creating new samples close to the existing minority instances, SMOTE increases the representation of these instances, thereby enhancing the classifier's generalization performance [35].

Specifically, for a minority class C to be augmented, we select a subset of nodes $\mathcal{V}_C$ from this class in training samples for SMOTE process. Initially, we choose a node v_i from $\mathcal{V}_C$ as an anchor node and identify its nearest neighbor v_j within the neighborhood as an auxiliary node. The SMOTE algorithm then generates a new node $\bar{v}$ by linearly interpolating the features of this node pair. This procedure is repeated for each node in the set $\mathcal{V}_C$, culminating in the augmentation of the fault samples. For node v_i and its neighbor v_j, the computation of feature $\bar{\mathbf{h}}$ of the synthesized node $\bar{v}$ is as follows:

$$\bar{\mathbf{h}} = \mathbf{h}_i + \lambda(\mathbf{h}_j - \mathbf{h}_i), \tag{10}$$

where $\bar{\mathbf{h}}$ represents the embedding representation of the synthesized minority class samples. $\mathbf{h}_i$ is a representation of a randomly selected minority class node v_i, $\mathbf{h}_j$ is one of the K nearest neighbors of node v_i within its class, $\lambda \in (0,1)$ is a hyperparameter responsible for controlling the proportion of the original node features in the newly synthesized nodes. Eventually, we denote the newly generated set of nodes as $\bar{\mathcal{V}}$, with the corresponding embedding representation matrix being $\bar{\mathbf{H}} \in \mathbb{R}^{\bar{n}\times d}$, where $\bar{n}$ and d respectively denote the number and dimensionality of the newly generated features.

Since the new nodes are synthesized within the minority class nodes, the label information of the newly synthesized nodes is consistent with the original nodes:

$$y_{\bar{v}} = y_{v_i} = y_{v_j}. \tag{11}$$

At this stage, we have obtained node representations and label information for the newly synthesized nodes. The application of the SMOTE method to each node in the minority class results in the generation of additional synthetic nodes. In the implementation, an over-sampling scale parameter δ is used to regulate the extent of new node generation. This approach helps in balancing the data volume across classes, thereby mitigating the adverse effects of biased models trained on imbalanced datasets concerning the minority class. Consequently, the augmented set of nodes is denoted as $\tilde{\mathcal{V}} = \{\mathcal{V} \cup \bar{\mathcal{V}}\}$, with the corresponding embedding matrix represented as $\tilde{\mathbf{H}} = [\mathbf{H} \parallel \bar{\mathbf{H}}]$.

Synthesizing Edges. Although we have generated features and labels for the synthesized nodes, these nodes remain isolated from the original graph. This disconnection limits our ability to leverage the graph structure to further assist the minority class. To address this issue, we develop an edge predictor to establish connections between the synthesized nodes and the original nodes. The enhanced graph and feature representations, including these newly established edges, are then input into the graph classifier for the final classification. In GNNs, feature information is propagated through edges, so we anticipate that the newly synthesized edges will effectively preserve the graph structure and improve classification performance. A practical solution is to design a component that generates edge connections compatible with the original graph structure. To this end, we introduce an edge predictor [36] that is trained on the original graph $\mathrm{G}^{(S)}$ and used to predict connections between synthesized nodes in $\bar{\mathcal{V}}$ and existing nodes in $\mathcal{V}$. Specifically, for nodes v_i and v_j, the edge predictor is implemented as follows:

$$\bar{\mathrm{A}}_{i,j} = \sigma(\mathbf{z}_i, \mathbf{z}_j^T); \mathbf{z}_i = \mathbf{W}_{syn}\mathbf{h}_i, \mathbf{z}_j = \mathbf{W}_{syn}\mathbf{h}_j, \tag{12}$$

where $\bar{\mathrm{A}}_{i,j}$ is used to predict the connectivity between nodes v_i and v_j in the original node set $\mathcal{V}$, $\mathbf{z}_i$ is obtained from $\mathbf{h}_i$ through a linear transformation, $\mathbf{W}_{syn}$ is a parameter matrix. The loss function for training the edge predictor is defined as follows:

$$\mathcal{L}_{edge} = \left\| \bar{\mathbf{A}} - \mathbf{A}^{(S)} \right\|_F^2. \tag{13}$$

With the edge predictor in place, we incorporate the predicted edges for the synthesized nodes into the augmented adjacency matrix. Specifically, the predicted probabilities between nodes are bounded by a threshold η:

$$\tilde{\mathrm{A}}_{\bar{v}_i, v_j} = \begin{cases} 1, & \bar{\mathrm{A}}_{\bar{v}_i, v_j} > \eta \\ 0, & \text{otherwise}, \end{cases} \tag{14}$$

where $\tilde{\mathrm{A}}_{\bar{v}_i, v_j}$ represents the connectivity between the synthesized node $\bar{v}_i$ and the original node v_j. Without loss of generality, we set the threshold η to 0.5. Thus, the connection between synthesized nodes and original nodes is established.

4.4. Graph classification and model optimization

Classification. Finally, we employ a GNN classifier to make the final predictions. For a node v_i:

$$\begin{aligned} \mathbf{h}_i^{(l+1)} &= \sigma(\tilde{\mathbf{W}}_1[\mathbf{h}_i^{(l)} \parallel \tilde{\mathbf{H}}^{(l)}\tilde{\mathbf{A}}[:,i]]), \\ \hat{y}_i &= \mathrm{Softmax}(\tilde{\mathbf{W}}_2\mathbf{h}_i^{(l+1)}), \end{aligned} \tag{15}$$

where $\tilde{\mathbf{W}}_1$ and $\tilde{\mathbf{W}}_2$ represent the weight parameters, $\hat{y}_i$ is the probability distribution of the predicted class labels for node v_i. The classifier module employs the cross-entropy loss function for optimization:

$$\mathcal{L}_{cls} = \text{Cross-Entropy}(\hat{\mathbf{Y}}, \mathbf{Y}^L). \tag{16}$$

Model Optimization. Combining the contrastive learning loss, edge predictor loss, and node classification loss, we formulate the final objective loss function of the model as follows:

$$\mathcal{L} = \mathcal{L}_{cls} + \mathcal{L}_{gcl} + \alpha\mathcal{L}_{edge}, \tag{17}$$

where $\mathcal{L}$ represents the overall training loss, α is a hyperparameter used to control the proportion of edge loss, with an initial value set to 0.001. We adopt a joint training approach [37], which simultaneously optimizes contrastive learning and classification loss.

4.5. Novelty discussion

Conventional DGA methods generally regard each sample as an independent entity, thereby making it impractical to aggregate information from other samples through propagation. To the best of our knowledge, this work is the first to apply a graph machine learning imbalance algorithm to transformer fault sample diagnosis. While previous works have made progress in this area, we believe our study differs in the following three aspects.

GNN backbone. We employ GNNs as the foundational architecture for our model to better capture the relationships between features, thereby providing more supervisory signals. In real-world scenarios, fault data are often sparse and difficult to obtain. Therefore, how to effectively extract distinguishable feature information from limited fault samples warrants thorough consideration. GNNs, due to their feature updating mechanism that incorporates information from neighboring nodes, are well-suited to handle such scenarios. Additionally, our experiments validate that the GNN architecture outperforms the Multi-Layer Perceptron (MLP) architecture in handling imbalanced fault data.

Data augmentation. We utilize data augmentation methods to enhance the minority class within the challenging-to-identify fault data. Specifically, we utilize the SMOTE interpolation method to generate minority class samples, thereby increasing feature richness and enabling the model to learn a wider variety of features. Furthermore, we establish connections between the newly generated samples and the original samples, harnessing the updating mechanisms of GNNs to further enhance model performance.

Dual-channel contrastive learning. We consider a contrastive learning approach on graphs to optimize the feature quality of fault samples learned by the GNNs. Unlike traditional contrastive learning, our method uses homophily and heterophily as guidance when constructing contrastive views. Specifically, a similar KNN graph is used to bring the representations of the same class fault nodes closer, while a dissimilar KNN graph is used to push the representations of different class fault nodes further apart. We apply contrastive learning to enforce consistency constraints on the embeddings under both views, and then integrate the embeddings from the two views in a balanced manner to achieve optimization.

4.6. Training algorithm

To clearly describe the overall training process of GraphSmin, we summarize the algorithm in **Algorithm** 1. Specifically, we first conduct (1) *preprocessing*: read and split the dataset, and construct similar and dissimilar KNN graphs based on feature attributes using the Gaussian Kernel Function (Line 2). (2) *Dual-channel contrastive learning*: the similarity and dissimilarity information obtained from the two perspectives is fed into a dual-channel filter for representation learning (Line 4). Simultaneously, we employ common contrastive enhancement strategies to optimize the effectiveness of contrastive learning (Line 5). (3) *Data generation and edge synthesis*: the model integrates fault sample embeddings learned from the dual perspectives and uses SMOTE to generate new data for minority class samples (Lines 6 and 7). Additionally, an edge predictor is utilized to generate connections between the original samples and the synthesized samples (Line 8). (4) *Fault sample classification*: the augmented samples, combined with the original samples, are then input into the classifier for classification (Line 9). Finally, we construct the overall loss function, which includes classification loss, contrastive loss, and edge predictor loss, to update the model parameters (Line 10).

Algorithm 1 Algorithm of **GraphSmin**

Input: Graph $\mathrm{G}^{(S)} = (\mathcal{V}, \mathbf{X}, \mathbf{A}^{(S)})$, $\mathrm{G}^{(NS)} = (\mathcal{V}, \mathbf{X}, \mathbf{A}^{(NS)})$, training labels $\mathbf{Y}^L$, number of classes C
Parameters: upsampling coefficient λ, imbalance coefficient ρ, feature mixing ratio β, edge predictor loss Ratio α
Output: Predicted labels $\mathbf{Y}^T$

1: Randomly initialize the feature encoder $\mathcal{F}_{en}$, edge predictor $\mathcal{F}_{edge}$ and node classifier $\mathcal{F}_{cls}$;
2: calculate similar graph $\mathrm{G}^{(S)}$ and dissimilar graph $\mathrm{G}^{(NS)}$ according to Eq.(3), (4) and (5).
3: **for** $epoch \leftarrow 1, 2, \cdots$ **do**
4: Calculate $\mathbf{H}^{(S)}$, $\mathbf{H}^{(NS)}$ via Eq. (6) and (7).
5: Compute $\mathcal{L}_{gcl}$ according to Eq. (8) and (9).
6: Acquire $\mathbf{H} = \beta\mathbf{H}^{(S)} + (1-\beta)\mathbf{H}^{(NS)}$.
7: Interpolation between node v_i and v_j in minority class to synthesize new nodes based on Eq. (10) and (11).
8: Generate $\tilde{\mathbf{A}}$ using edge predictor based on Eq. (12), (13) and (14).
9: Update the model by minimizing $\mathcal{L}_{cls} + \mathcal{L}_{gcl} + \alpha\mathcal{L}_{edge}$ with standard backpropagation.
10: Update the parameters in the model with the Adam optimizer.
11: **return** $\mathcal{F}_{cls}$

5. Experiments

In this section, we conduct extensive experiments to evaluate the effectiveness of our framework and aim to address the following research questions:

- **RQ1:** How does GraphSmin perform compared to other baseline methods in DGA under different imbalance scenarios?
- **RQ2:** How does different components of GraphSmin contribute to performance improvement?
- **RQ3:** Is GraphSmin sensitive to hyperparameters?
- **RQ4:** How can we intuitively understand the accuracy of diagnosing minority class faults?

5.1. Experiment settings

5.1.1. Datasets

We utilize two publicly available datasets named DGA-200[1] and DGA-2000[2] for model construction and performance evaluation. The first dataset comprises 1982 DGA samples, while the second contains 201 DGA samples. The statistical information of these datasets is provided in Table 1.

DGA-200: The DGA-200 dataset consists of 201 samples, categorized into seven fault types: Partial Discharge (PD), Spark Discharge (SD), Arc Discharge (AD), Low-Temperature Overheating (LT), Low to Medium Temperature Overheating (LMT), Medium Temperature Overheating (MT), and High Temperature Overheating (HT). Each sample

[1] https://ieee-dataport.org/documents/dissolved-gas-data-transformer-oil-fault-diagnosis-power-transformers-membership-degree.
[2] https://github.com/alan-456/transformer-fault-dataset.

includes information on five types of dissolved gases: H_2 (Hydrogen), CH_4(Methane), C_2H_6(Ethane), C_2H_4(Ethylene), and C_2H_2(Acetylene).

DGA-2000: The DGA-2000 dataset comprises 1982 samples, including six fault categories: Low-Temperature Overheating (LT), Medium-Temperature Overheating (MT), High-Temperature Overheating (HT), High-Energy Discharge (HE), Low-Energy Discharge (LE), Partial Discharge (PD), and one normal category (Normal). The types of gases in the data are consistent with those in DGA-200. Moreover, both datasets exhibit characteristics of imbalanced distributions. For instance, in DGA-2000, the ratio between the largest and smallest class sample counts reached 6.26.

5.1.2. Baselines

To validate the performance of our framework, we compare it with two groups of representative methods, including traditional machine learning algorithms and machine learning algorithms with data augmentation. The first group of methods consists of general machine learning algorithms, while the second group is specifically designed for imbalanced scenarios.

(1) Traditional machine learning algorithms:

- **SVM**: Support Vector Machine is a supervised learning algorithm used for classification and regression tasks. Its core idea is to find an optimal hyperplane that separates data points of different classes and maximizes the margin between two classes. SVM is highly effective in handling high-dimensional and nonlinear data, and it can address complex classification problems through kernel tricks.
- **DTree**: Decision Tree algorithm is a common machine learning algorithm used for classification and regression tasks. It constructs a tree-like structure to represent decision rules, where each node represents a feature test, and each leaf node represents a class label or regression value. The algorithm builds the tree structure by selecting the best features for splitting to classify or predict samples.
- ***K*NN**: K-nearest neighbors algorithm is a non-parametric method used for pattern recognition and regression analysis. The core idea of this algorithm is sample-based learning, where classification or regression predictions are made by finding the K nearest neighbors in the training dataset that are most similar to the test sample. In classification tasks, KNN determines the class of the test sample through a voting mechanism, while in regression tasks, it predicts the value of the target variable by computing the average of the K nearest neighbors.

(2) Machine learning algorithms with SMOTE data augmentation:

- **SVM-SMOTE**: First, we apply the SMOTE algorithm to augment minority class fault samples, then perform SVM classification.
- **DTree-SMOTE**: We Apply the SMOTE algorithm to augment minority class fault samples firstly, then perform Decision Tree classification.
- ***K*NN-SMOTE**: This is consistent with the previous two algorithms.

Additionally, to investigate the performance gains brought by GNN aggregation operations, we implement a variant of GraphSmin called:

- **GraphSmin**$_{\text{MLP}}$: It adopts a backbone based on MLP. Specifically, encoding and classification processes are implemented based on MLP rather than GNN structures.

Below are our models under two different settings, each utilizing a different GNN architecture backbone for feature aggregation and classification:

- **GraphSmin**$_1$: This is Similar to the GraphSmin$_{MLP}$ setting, but with a GCN backbone.
- **GraphSmin**$_2$: We replace the backbone of GraphSmin with GraphSage.

Table 2
Hyper-parameter settings for GraphSmin.

Parameter	DGA-2000	DGA-200
α	(0.001,0.01)	(0.001,0.01)
K	$\{2, 4, 8, 20, 30\}$	$\{1, 2, 4, 5, 10\}$
ρ	$\{2, 5, 10, 20\}$	$\{2, 4, 5, 10\}$
Epochs	1500	1000
β	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$

5.1.3. Evaluation metrics

ACC: Accuracy is a metric used to evaluate the performance of classification models. It measures the proportion of correctly classified samples, i.e., the ratio of the number of samples correctly classified to the total number of samples. A higher ACC indicates higher classification accuracy and is one of the commonly used metrics for evaluating classification performance.

AUC-ROC: Area Under the Receiver Operating Characteristic Curve (AUC-ROC) is a metric used to evaluate the performance of binary classification models. It measures performance of the model at different classification thresholds and represents the area under the ROC curve. AUC-ROC closer to 1 indicates better model performance, while 0.5 indicates performance equivalent to random prediction.

F1 Score: F1 score is a comprehensive metric for evaluating the performance of classification models. It combines precision and recall of the model to balance the impact of these two metrics. A higher F1 score indicates a better balance between precision and recall and is a commonly used metric for evaluating classification performance.

5.1.4. Parameter settings

We implement our proposed GraphSmin along with some necessary baseline models. For SVM, KNN, and DTree, we utilize corresponding library functions and determine the optimal hyperparameter settings using grid search. For GraphSmin, we adjust the following hyperparameters: when constructing the graph structure, the number of nearest neighbors $K \in \{2, 3, 4, 5, 10, 15, 20, 30\}$, dropout rates for contrastive learning of nodes and edges set to 0.1 and 0.5 respectively, edge generation loss coefficient $\alpha \in (0.001, 0.01)$, imbalance ratio $\rho \in (2, 40)$, up-sampling ratio $\lambda \in (0.1, 1)$ and training $epochs \in (1500, 2000, 2500)$. We select two-layer GCN and GraphSage with 128 hidden units as encoders, and the dimensionality of decoder units is set to 128. Additionally, we have tabulated the key parameters in Table 2 for comprehensive reference.

5.2. Performance comparison (RQ1)

In this subsection, we compare the performance of all the methods under different data partitions, following the common imbalanced setting, i.e., Long-tailed Sampling and Step Sampling. The results are summarized in Tables 3 and 4. Note that all results are averaged over 10 runs and accompanied by their standard deviations. The best method is indicated in bold, while the runner-up method is underlined.

Long-tailed Sampling. Long-tailed Sampling involves sampling the training set according to an exponentially decaying rule based on the number of samples in each class. Specifically, we fix the number of majority class samples and gradually reduce the number of minority class samples according to a decay factor to meet the specified imbalance ratio. In Table 4, we summarize the following observations: **(1)** Compared to traditional machine learning algorithms such as SVM, KNN, and DTree, our approach achieves the best performance under both datasets, with significant advantages. For example, our method achieves performance improvements of 11.19% and 7.46% in AUC and Macro-F1 scores, respectively, on the DGA-2000 dataset compared to SVM. **(2)** Machine learning algorithms with data augmentation achieve performance improvements on almost all three evaluation metrics for both datasets, further validating the effectiveness of our framework

Table 3
Long-tailed sampling.

Dataset	DGA-2000			DGA-200		
Imb ratio	5			5		
Eval metric	Acc	AUC-ROC	F1	Acc	AUC-ROC	F1
SVM	0.6886 ± 0.0281	0.8183 ± 0.0164	0.6672 ± 0.0352	0.5743 ± 0.0657	0.7517 ± 0.0383	0.5381 ± 0.0788
Dtree	0.6086 ± 0.0443	0.7717 ± 0.0258	0.5839 ± 0.0486	0.5943 ± 0.0698	0.7633 ± 0.0407	0.5666 ± 0.0737
*K*NN	0.6833 ± 0.0235	0.8153 ± 0.0137	0.6623 ± 0.0208	0.5686 ± 0.0871	0.7483 ± 0.0508	0.5483 ± 0.0989
SVM-SMOTE	0.6771 ± 0.0201	0.8117 ± 0.0117	0.6666 ± 0.0215	0.6086 ± 0.0798	0.7717 ± 0.0466	0.5867 ± 0.0938
Dtree-SMOTE	0.6338 ± 0.0283	0.7864 ± 0.0165	0.6235 ± 0.0286	0.6114 ± 0.0966	0.7733 ± 0.0564	0.5920 ± 0.1009
*K*NN-SMOTE	0.7090 ± 0.0198	0.8303 ± 0.0116	0.7013 ± 0.0207	0.5886 ± 0.0574	0.7600 ± 0.0335	0.5742 ± 0.0573
GraphSmin$_{MLP}$	0.7290 ± 0.0339	0.9186 ± 0.0096	0.7213 ± 0.0354	**0.7057 ± 0.0558**	**0.9160 ± 0.0243**	0.6971 ± 0.0608
GraphSmin$_1$	**0.7486 ± 0.0277**	0.9272 ± 0.0116	**0.7418 ± 0.0276**	0.6771 ± 0.0479	0.9003 ± 0.0208	0.6701 ± 0.0493
GraphSmin$_2$	0.7476 ± 0.0307	**0.9302 ± 0.0095**	0.7411 ± 0.0316	0.7029 ± 0.0514	0.9105 ± 0.0238	**0.6973 ± 0.0524**

Table 4
Step sampling.

Dataset	DGA-2000			DGA-200		
Imb ratio	5			5		
Eval metric	Acc	AUC-ROC	F1	Acc	AUC-ROC	F1
SVM	0.6695–0.0358	0.8072–0.0209	0.6651 ± 0.0377	0.4821 ± 0.0559	0.6979 ± 0.0326	0.4152 ± 0.0634
Dtree	0.5871 ± 0.0256	0.7592 ± 0.0150	0.5788 ± 0.0327	0.5214 ± 0.0863	0.7208 ± 0.0503	0.4710 ± 0.1063
*K*NN	0.6738 ± 0.0453	0.8097 ± 0.0264	0.6662 ± 0.0499	0.4643 ± 0.0714	0.6875 ± 0.0417	0.4022 ± 0.0891
SVM-SMOTE	0.6690 ± 0.0362	0.8069 ± 0.0211	0.6633 ± 0.0389	0.5679 ± 0.0586	0.7479 ± 0.0342	0.5483 ± 0.0657
Dtree-SMOTE	0.5919 ± 0.0429	0.7619 ± 0.0250	0.5888 ± 0.0410	0.5286 ± 0.1150	0.7250 ± 0.0671	0.5003 ± 0.1203
*K*NN-SMOTE	0.6881 ± 0.0385	0.8181 ± 0.0225	0.6819 ± 0.0405	0.5321 ± 0.0667	0.7271 ± 0.0389	0.5101 ± 0.0798
GraphSmin$_{MLP}$	0.7195 ± 0.0284	0.9170 ± 0.0100	0.7156 ± 0.0283	**0.6357 ± 0.0674**	**0.8946 ± 0.0389**	**0.6172 ± 0.0666**
GraphSmin$_1$	**0.7267 ± 0.0296**	0.9256 ± 0.0099	**0.7234 ± 0.0306**	0.6143 ± 0.0614	0.8778 ± 0.0351	0.5954 ± 0.0697
GraphSmin$_2$	0.7248 ± 0.0267	**0.9276 ± 0.0094**	0.7221 ± 0.0269	0.6250 ± 0.0486	0.8774 ± 0.0459	0.6096 ± 0.0507

in using SMOTE to augment minority class samples. However, the accuracy of SVM slightly decreases after data augmentation on the DGA-2000 dataset, possibly due to the noise introduced by the augmented samples interfering with discrimination. **(3)** GraphSmin$_{MLP}$ exhibit suboptimal performance on the DGA-2000 dataset but achieve near-optimal performance on the DGA-200 dataset. This phenomenon indicates that the superiority of GNNs methods depends on a reasonable graph structure foundation [38], we will leave a more detailed discussion and design of graph construction methods as further work in the future.

Step Sampling. Another imbalance handling method is Step Sampling. Without loss of generality, we adopt the default approach, namely considering the last three classes of fault samples in the minority classes as the classes to be augmented. Table 3 summarizes the experimental results on the two datasets with an imbalance ratio of 5, and our observations are as follows: **(1)** In the vast majority of cases, the GraphSmin framework achieves the best performance. Using the GCN framework yields even better performance compared to the Sage framework. **(2)** Similar to the results under the Long-tailed Sampling setting, machine learning algorithms with SMOTE show performance improvements on both datasets, further confirming the effectiveness of using SMOTE for data augmentation. **(3)** GraphSmin$_{MLP}$ achieve the best performance on the DGA-200 dataset, reflecting the performance gains brought by a reasonable graph structure, particularly the higher node homophily.

5.3. Ablation study (RQ2)

To assess the effectiveness of each component, we conduct several ablation experiments on DGA-2000 dataset, applying the GCN backbone and Long-tailed Sampling setting. We compare GraphSmin with its three variants to analyze the effectiveness of each module as follows:

- **w/oGCL**: GraphSmin without graph contrastive learning.
- **w/oSMOTE**: GraphSmin without data augmentation.
- **w/oDecoder**: GraphSmin without edge structure generation.
- **GraphSmin(MLP)**: Replace the GNN-based encoder with the MLP-based encoder.
- **GraphSmin**: The complete model with all components utilized.

We report the average ACC, AUC, and Macro-F1 over 10 runs under the Long-tailed sampling setting with ρ equal to 5. From the results in Fig. 6, we can draw the following conclusions: **(1)** GraphSmin outperforms all variants, indicating the effectiveness of each component in enhancing model performance. **(2)** The structural relationships between newly generated sample nodes from decoder and original nodes contributed to better classification, as evidenced by the larger discrepancies observed when the Decoder module is ablated compared to other components. **(3)** Contrastive learning improves embedding quality, we attribute to the design of the dual-view approach and the construction of contrastive loss. The performance of the model decreases by 1.19% in ACC and 1.47% in Macro-F1 when the contrastive learning component is disabled compared to GraphSmin. (4) When the model's encoder is replaced from GNN to MLP, the performance becomes suboptimal due to the lack of additional supervision provided by the neighbors. This further highlights the advantages of GNN-based methods. Additionally, the significant performance gains from SMOTE data augmentation underscore the benefits of adding minority class samples.

5.4. Parameter analysis (RQ3)

To rigorously evaluate the impact of different parameters on the model, we study and analyze the sensitivity of hyperparameters in the GraphSmin model on the DGA-2000 dataset.

Analysis of the imbalance coefficient ρ: ρ is used to control the imbalance ratio of the data, i.e., the ratio of the number of majority class samples to minority class samples. Observing Fig. 7 (left), it can be seen that as the imbalance ratio ρ increases, the proportion of minority class samples diminishes in the overall data, and the model performance gradually decreases. However, our method still performs significantly better than the baseline models. Additionally, our model demonstrates smaller errors and higher stability.

Analysis of the feature mixing coefficient β: Different feature mixing coefficients enable the acquisition of embedding information

Table 5
CASE study of accuracy for each class on DGA-2000 dataset.

Class(Types) Distribution	C_0 (LT) 3.41%	C_1 (MT) 5.12%	C_2 (PD) 7.51%	C_3 (HE) 10.91%	C_4 (LE) 15.70%	C_5 (HT) 23.21%	C_6 (Normal) 34.13%
SVM	0.1167 ± 0.0847	0.6467 ± 0.0777	0.8033 ± 0.0314	0.5433 ± 0.1212	0.7767 ± 0.0700	0.8733 ± 0.0867	0.9200 ± 0.0371
*K*NN	0.0633 ± 0.0657	0.5800 ± 0.0833	0.7667 ± 0.0615	0.5067 ± 0.0490	0.7433 ± 0.0667	**0.9033 ± 0.0657**	**0.9267 ± 0.0359**
DTree	0.1167 ± 0.0687	0.4833 ± 0.1310	0.6400 ± 0.1123	0.5200 ± 0.1118	0.7300 ± 0.1149	0.7900 ± 0.0907	0.9133 ± 0.0653
SVM-SMOTE	0.2433 ± 0.0989	0.6700 ± 0.0948	0.7900 ± 0.0517	0.5167 ± 0.1258	0.7633 ± 0.1016	0.8500 ± 0.0654	0.9167 ± 0.0500
*K*NN-SMOTE	0.2100 ± 0.0920	0.6867 ± 0.0542	0.7833 ± 0.0543	0.4967 ± 0.0674	0.7333 ± 0.0856	0.8633 ± 0.0722	0.9100 ± 0.0473
Dtree-SMOTE	0.2233 ± 0.0967	0.5500 ± 0.0719	0.6433 ± 0.0895	0.5033 ± 0.1140	0.6733 ± 0.1397	0.7533 ± 0.0968	0.8933 ± 0.0593
GraphSmin_1	**0.4400 ± 0.1267**	**0.7133 ± 0.0618**	**0.8467 ± 0.0427**	**0.6233 ± 0.0790**	0.7633 ± 0.0964	0.8833 ± 0.0593	0.9067 ± 0.0636
GraphSmin_2	0.3800 ± 0.0746	0.7000 ± 0.0448	0.8233 ± 0.0400	0.5933 ± 0.0892	**0.8033 ± 0.0836**	0.8900 ± 0.0651	0.9167 ± 0.0512

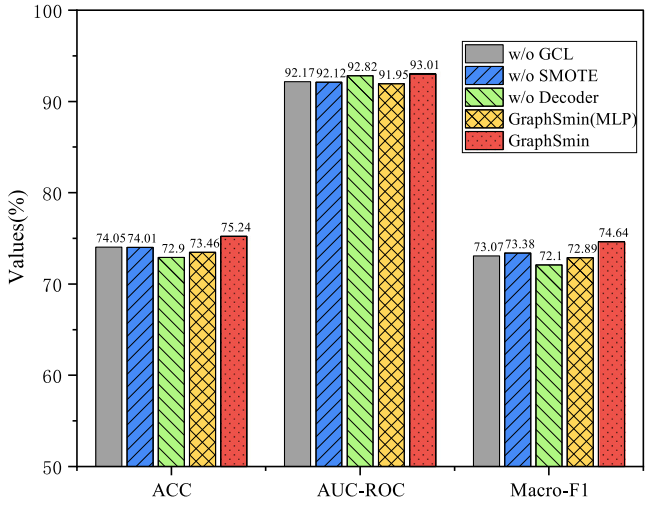


Fig. 6. Ablation study on DGA-2000.

from dual perspectives in varying proportions. As observed in Fig. 7 (right), when the embedding proportion from the homophily perspective ranges from 0.6 to 0.7, the model achieves optimal performance. However, higher or lower proportions have an impact on the model, which indicates that an appropriate proportion of embeddings from the heterophily perspective brings information gain, thereby promoting performance improvement. Excessive or insufficient message may introduce noise or insufficient information, neither of which allows the model to reach its optimum. The conclusion further confirms that the information gain from dual-perspective contrastive learning promotes performance enhancement.

Analysis of the number of neighbors K: The value of K is used to control the number of neighbors selected when constructing the *K*NN graph, resulting in different graph structures and thus different results. We adopt Long-tailed Sampling and report the average AUC-ROC and Macro-F1 scores over 10 runs under different values of K, as shown in Fig. 8. It can be observed that smaller values of K exhibit superior results, while larger values of K lead to decreased model performance. However, this experimental result contradicts intuition, as based on the assumption of graph homophily, neighbors are expected to provide more supervisory signals, thereby facilitating performance improvement. To address this, we compare the homophilic scores of graphs constructed under different values of K, as well as the homophilic scores of each subclass. The results reveal that the graph constructed from DGA-2000 achieves the highest homophilic score when K equals 4 or 5. Higher homophilic scores indicate that neighbors are more likely to belong to the same class, thereby providing beneficial information. Conversely, lower homophilic scores suggest that non-similar neighbors introduce noise signals through aggregation, and the more neighbors, the greater the cumulative noise. Hence, the phenomenon of decreasing performance with larger values occurs. Therefore, a reasonable graph structure is crucial for GNNs methods.

5.5. Case study (RQ4)

We conduct a case study on the accuracy differences between the baseline and GraphSmin for each class on the DGA-2000 dataset under the Long-tailed Sampling setting. The results are summarized in Table 5. Analysis of the table yields the following observations: **(1)** The main issue with traditional machine learning lies in its inability to effectively handle minority class fault data, thereby limiting its overall performance. For instance, the *K*NN algorithm exhibits accuracy as low as 6.33% for class C_0, while achieving optimal accuracy of 92.67% for majority class, such as class C_6. Similarly, other baseline models show similar results, reflecting the trade-off between recognizing majority class fault data and sacrificing minority class performance. **(2)** Machine learning algorithms based on SMOTE to some extent improve the recognition accuracy of minority class fault data, indicating the effectiveness of SMOTE in augmenting minority class samples. However, there is a decrease in the performance of majority classes, which attribute to the interference of closely generated samples with the judgment of the baseline model. **(3)** Compared to the baselines, our method significantly improves the performance of identifying minority class fault samples while ensuring the performance of majority classes. This improvement is attributed to the additional information gain provided by the graph structure through neighbors.

6. Conclusion

In this study, we introduce a novel framework named GraphSmin, designed to elevate the performance of the model in recognizing fault samples of dissolved gases in oil. Concretely, the SMOTE algorithm is employed to dynamically synthesize minority class fault samples, enabling the model to focus more on the minority class. Concurrently, an edge predictor is utilized to establish the connection between synthetic fault samples and original samples. Furthermore, we contemplate harnessing contrastive learning to refine node representations of fault samples under dual perspectives. Our thorough experimental assessment showcases remarkable ability of GraphSmin to classify minority classes while preserving accuracy in identifying majority classes under both Long-tailed and Step Sampling settings, which is crucial for various practical applications. For future work, we aim to investigate how to enhance the quality of graphs for accelerating downstream tasks in imbalanced DGA scenarios.

CRediT authorship contribution statement

Ke Shu: Writing – original draft, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Huifang Ma:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation, Conceptualization. **Jinpeng Yang:** Resources, Methodology, Formal analysis, Data curation. **Di Zhang:** Supervision, Resources, Formal analysis.

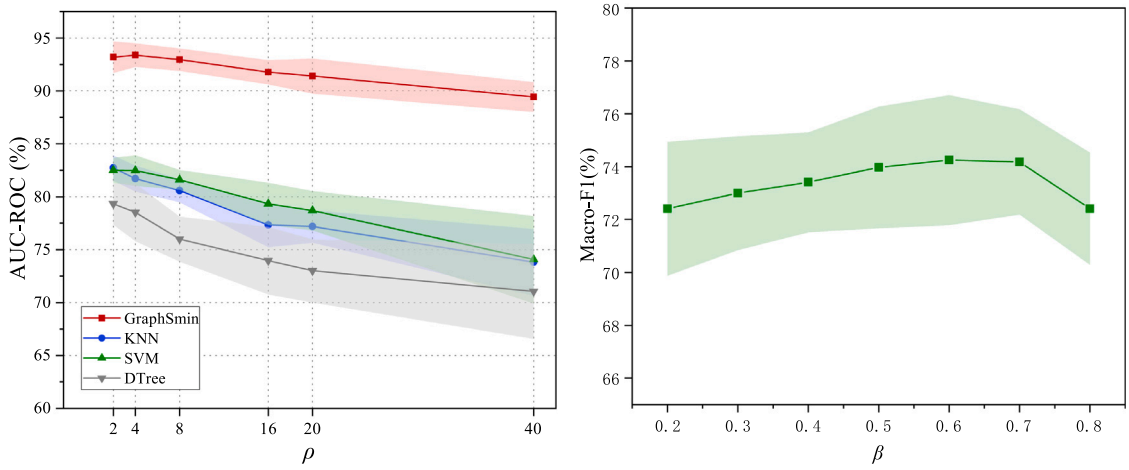


Fig. 7. Parameter sensitivity analysis on DGA-2000 with different ρ and β.

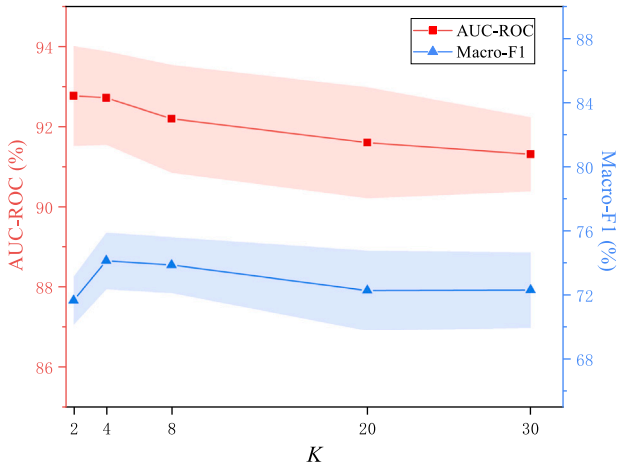


Fig. 8. Parameter sensitivity analysis on DGA-2000 under different K-nearest neighbors.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: We would like to submit the manuscript entitled 'GraphSmin: Imbalanced Dissolved Gas Analysis with Contrastive Dual-Channel Graph Filters', which we wish to be considered for publication in Advanced Engineering Informatics. No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication.

Data availability

I have open-sourced the code and data on GitHub, and added the links to the abstract of the article.

Acknowledgments

This study is supported by the Industrial Support Project of Gansu Colleges, China (No. 2022CYZC11)

References

[1] L. Hong, Z. Chen, Y. Wang, M. Shahidehpour, M. Wu, A novel SVM-based decision framework considering feature distribution for Power Transformer Fault Diagnosis, Energy Rep. 8 (2022) 9392–9401.

[2] W. Zhang, J. Yuan, T. Zhang, K. Zhang, An improved three-ratio method for transformer fault diagnosis using B-spline theory, Proc. CSEE 34 (2014) 4129–4136.

[3] Y. Zhang, H. Wei, R. Liao, Y. Wang, L. Yang, C. Yan, A new support vector machine model based on improved imperialist competitive algorithm for fault diagnosis of oil-immersed transformers, J. Electr. Eng. Technol. 12 (2) (2017) 830–839.

[4] Y. Zhang, H.C. Chen, Y. Du, M. Chen, J. Liang, J. Li, X. Fan, X. Yao, Power transformer fault diagnosis considering data imbalance and data set fusion, High Volt. 6 (3) (2021) 543–554.

[5] N.A. Bakar, A. Abu-Siada, S. Islam, A review of dissolved gas analysis measurement and interpretation techniques, IEEE Electr. Insul. Mag. 30 (3) (2014) 39–49.

[6] S.A. Wani, A.S. Rana, S. Sohail, O. Rahman, S. Parveen, S.A. Khan, Advances in DGA based condition monitoring of transformers: A review, Renew. Sustain. Energy Rev. 149 (2021) 111347.

[7] M. Demirci, H. Gözde, M.C. Taplamacioglu, Improvement of power transformer fault diagnosis by using sequential Kalman filter sensor fusion, Int. J. Electr. Power Energy Syst. 149 (2023) 109038.

[8] X. Tan, J. Qi, J.Q. Gan, J. Zhang, C. Guo, F. Wan, K. Wang, Multi-filter semi-supervised transformer model for fault diagnosis, Eng. Appl. Artif. Intell. 124 (2023) 106498.

[9] J.I. Aizpurua, V.M. Catterson, B.G. Stewart, S.D. McArthur, B. Lambert, B. Ampofo, G. Pereira, J.G. Cross, Power transformer dissolved gas analysis through Bayesian networks and hypothesis testing, IEEE Trans. Dielectr. Electr. Insul. 25 (2) (2018) 494–506.

[10] M. Sewwandi, Y. Li, J. Zhang, A class-specific feature selection and classification approach using neighborhood rough set and K-nearest neighbor theories, Appl. Soft Comput. 143 (2023) 110366.

[11] B. Charbuty, A. Abdulazeez, Classification based on decision tree algorithm for machine learning, J. Appl. Sci. Technol. Trends 2 (01) (2021) 20–28.

[12] A. Roy, S. Chakraborty, Support vector machine in structural reliability analysis: A review, Reliab. Eng. Syst. Saf. 233 (2023) 109126.

[13] W. Jin, H. Ma, Y. Zhang, Z. Li, L. Chang, Multi-view discriminative edge heterophily contrastive learning network for attributed graph anomaly detection, Expert Systems with Applications (2024) 124460.

[14] Y. Ma, Y. Tian, N. Moniz, N.V. Chawla, Class-imbalanced learning on graphs: A survey, 2023, arXiv preprint arXiv:2304.04300.

[15] Y. Zhang, B. Kang, B. Hooi, S. Yan, J. Feng, Deep long-tailed learning: A survey, IEEE Trans. Pattern Anal. Mach. Intell. (2023).

[16] J. Fan, Z. Liu, A. Meng, H. Yin, Q. Sun, F. Bin, Q. Jiang, Characteristics of tin oxide chromatographic detector for dissolved gases analysis of transformer oil, IEEE Access 7 (2019) 94012–94020.

[17] Y. Li, Y. Xu, X. Li, R. Li, J. Lin, G. Zhang, Addressing imbalance of sample datasets in dissolved gas analysis by data augmentation: Generative adversarial networks, IET Gener. Transm. Distrib. 16 (22) (2022) 4494–4504.

[18] S.M. de Andrade Lopes, R.A. Flauzino, R.A.C. Altafim, Incipient fault diagnosis in power transformers by data-driven models with over-sampled dataset, Electr. Power Syst. Res. 201 (2021) 107519.

[19] G. Haixiang, L. Yijing, J. Shang, G. Mingyun, H. Yuanyue, G. Bing, Learning from class-imbalanced data: Review of methods and applications, Expert Syst. Appl. 73 (2017) 220–239.

[20] Z. Liu, Y. Li, N. Chen, Q. Wang, B. Hooi, B. He, A survey of imbalanced learning on graphs: Problems, techniques, and future directions, 2023, arXiv preprint arXiv:2308.13821.

[21] S. Rezvani, X. Wang, A broad review on class imbalance learning techniques, Appl. Soft Comput. (2023) 110415.
[22] K. Xu, W. Hu, J. Leskovec, S. Jegelka, How powerful are graph neural networks? 2018, arXiv preprint arXiv:1810.00826.
[23] T.N. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, 2016, arXiv preprint arXiv:1609.02907.
[24] H. Zhang, B. Wu, X. Yuan, S. Pan, H. Tong, J. Pei, Trustworthy graph neural networks: Aspects, methods, and trends, Proc. IEEE (2024).
[25] Y. Liu, M. Jin, S. Pan, C. Zhou, Y. Zheng, F. Xia, S.Y. Philip, Graph self-supervised learning: A survey, IEEE Trans. Knowl. Data Eng. 35 (6) (2022) 5879–5900.
[26] Y. Liu, Y. Zheng, D. Zhang, V.C. Lee, S. Pan, Beyond smoothing: Unsupervised graph representation learning with edge heterophily discriminating, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 37, 2023, pp. 4516–4524.
[27] Y. Zhu, Y. Xu, F. Yu, Q. Liu, S. Wu, L. Wang, Graph contrastive learning with adaptive augmentation, in: Proceedings of the Web Conference 2021, 2021, pp. 2069–2080.
[28] W. Hamilton, Z. Ying, J. Leskovec, Inductive representation learning on large graphs, Adv. Neural Inf. Process. Syst. 30 (2017).
[29] J. Huang, L. Du, X. Chen, Q. Fu, S. Han, D. Zhang, Robust mid-pass filtering graph convolutional networks, in: Proceedings of the ACM Web Conference 2023, 2023, pp. 328–338.
[30] N. Keriven, Not too little, not too much: a theoretical analysis of graph (over) smoothing, Adv. Neural Inf. Process. Syst. 35 (2022) 2268–2281.
[31] J. Zhu, R.A. Rossi, A. Rao, T. Mai, N. Lipka, N.K. Ahmed, D. Koutra, Graph neural networks with heterophily, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 35, 2021, pp. 11168–11176.
[32] K. He, H. Fan, Y. Wu, S. Xie, R. Girshick, Momentum contrast for unsupervised visual representation learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 9729–9738.
[33] Y. Zhu, Y. Xu, F. Yu, Q. Liu, S. Wu, L. Wang, Deep graph contrastive representation learning, 2020, arXiv preprint arXiv:2006.04131.
[34] Z. Wu, H. Ma, B. Deng, Z. Li, L. Chang, Dual-channel dual-scale interactive learning for the prediction of compound-protein interaction, in: International Conference on Database Systems for Advanced Applications, Springer, 2024, pp. 155–171.
[35] A. Fernández, S. Garcia, F. Herrera, N.V. Chawla, SMOTE for learning from imbalanced data: progress and challenges, marking the 15-year anniversary, J. Artif. Intell. Res. 61 (2018) 863–905.
[36] T. Zhao, X. Zhang, S. Wang, Graphsmote: Imbalanced node classification on graphs with graph neural networks, in: Proceedings of the 14th ACM International Conference on Web Search and Data Mining, 2021, pp. 833–841.
[37] K. Ding, Z. Xu, H. Tong, H. Liu, Data augmentation for deep graph learning: A survey, ACM SIGKDD Explor. Newsl. 24 (2) (2022) 61–77.
[38] H. Mao, Z. Chen, W. Jin, H. Han, Y. Ma, T. Zhao, N. Shah, J. Tang, Demystifying structural disparity in graph neural networks: Can one size fit all? Adv. Neural Inf. Process. Syst. 36 (2024).